**DATA STRUCTURES AND ALGORITHMS** **SPRING 2026**

**ASSIGNMENT 04** **TOTAL MARKS: 10**

# CREATING A BINARY SEARCH TREE

**PROBLEM STATEMENT**

1. Create a Binary Search Tree using the Base Variant. Following functions are required:
    a. Insert a Node
    b. Delete a Node
    *(Cater all three cases, Leaf node, Node with one child and a Node with two children)*
    c. Search a Key provided by the user
    d. A function to add numbers to a queue reading the tree in-order
    e. Find the Maximum Key in the tree
    *(Hint: Use the extreme right node in the right subtree)*
    f. Find the Minimum Key in the tree
    *(Hint: Use the extreme left node in the left subtree)*

***Following are the requirements:***

Assignment must be submitted in hard copy and hand written. You need to write the same code five times at least. Use alternate ink (**Blue** and **Red**) **2 marks each**